

COMPARISON REPORT

VALIDATION TOOL	DESCRIPTION	BEFORE	AFTER	CHANGE
ERRAT	Overall Quality Factor	97.9592	98.9247	+0.9655 (Improved)
PROCHECK	Errors	3	2	-1 (Improved)
	Warning	3	1	-2 (Improved)
	Pass	2	5	+3 (Improved)
Verify3D	% of Residues with Average 3D/1D Score ≥ 0.1	71.17	72.07	0.9 (Improved)

The protein structure (PDB ID: 1JSG) was validated using ERRAT, PROCHECK, and Verify3D. The model was refined and the refined structure re-validated using the same tools. Both validated results of the pre-refined protein and post-refined protein were then compared. From the results, the majority of the quality indicators showed marked improvement, whereas Verify3D remained below its recommended threshold.

- 1. ERRAT:** Overall Quality Factor increased from 97.9592 to 98.9247. ERRAT scores closer to 100 indicates a good structural quality and this means the protein structure was more favorable after refinement.
- 2. PROCHECK:** Errors decreased from 3 to 2, warning decreased from 3 to 1, and pass increased from 2 to 5. From the results, it is evident that the overall evaluation markedly improved.
- 3. Verify3D:** Percentage of residues with average 3D/1D score ≥ 0.1 slightly improved from 71.17 to 72.07. Because the recommended threshold is 80%, both validated pre-refined and post-refined structure remained classified as **FAIL**. Despite the improvement, the model structure still has limitations in residue-environment compatibility.

In conclusion, the model structure improved markedly, and the post-validation of the refined structure showed a better-quality profile than the pre-validated model. However, there may be a need for further refinement due to the persistent Verify3D failure.